

NYC Taxi Data — Visualization & Analysis Summary

Python DA Assignment 2 · Dataset: seaborn *taxis* (6,433 trips; 6,407 after cleaning)

Data Cleaning

The dataset had missing values in **payment**, **pickup_zone**, **pickup_borough**, **dropoff_zone** and **dropoff_borough**. Payment and dropoff fields were filled with the column mode; rows missing a pickup location were dropped since location drives most of the analysis. No numeric columns had missing values. Timestamps were converted to datetime. The cleaned dataset has 6,407 rows.

Key Findings

- **Manhattan dominates.** It accounts for the large majority of total fare revenue (~\$58,753 of ~\$83,500 across all boroughs) and the highest trip count, far ahead of Queens (~\$16,382), Brooklyn (~\$6,327) and the Bronx (~\$2,079).
- **Trips are mostly short.** Median distance is 1.65 miles and 75% of trips are under 3.2 miles, giving a right-skewed distance histogram with a long tail of longer trips (max ~36.7 mi).
- **Distance and fare are strongly correlated** ($r = 0.94$), and both correlate closely with total ($r = 0.92$ and 0.97) — fare is the main driver of the total charge, more than tolls ($r = 0.61$ – 0.68) or tip ($r = 0.46$ – 0.63).
- **Credit card is the dominant payment method**, used in about 72% of trips versus 28% cash. Average fare is also higher on credit card (\$13.60) than cash (\$11.61) — this gap partly reflects that only credit-card trips have recorded tips in this dataset, so cash-trip tips read as \$0.
- **Tips are concentrated in Manhattan.** The median tip is about \$1.86 in Manhattan versus \$0.00 in the Bronx, Brooklyn and Queens — most tips outside Manhattan are logged as zero, likely reflecting the cash-tip recording gap combined with lower trip volume in those boroughs.
- **Fare distribution by payment method** (violin plot) is broadly similar in shape between cash and credit card, with credit card showing a slightly heavier upper tail of high-fare trips.

Visualizations Produced

Library	Chart	What it shows
Matplotlib/Pandas	Line	Fare over time (by pickup timestamp)
	Bar	Total fare by pickup borough
	Pie	Share of trips by payment method
	Histogram	Distribution of trip distance
	Box plot	Tip amount by pickup borough
Seaborn	Count plot	Number of trips by pickup borough
	Scatter plot	Distance vs. fare, colored by borough
	Heatmap	Correlation between distance, fare, tip, tolls, total
	Pair plot	Pairwise relationships (top 5 pickup zones)
	Violin plot	Fare distribution by payment method

Note: the pair plot colors only the 5 most frequent pickup zones (out of 100+) to keep the legend readable; all other charts use the full cleaned dataset.